

CONCORD v0.3 Scan Output Schemas

Normative scan artifacts for readiness, danger, and dependency analysis

Status: Normative draft | **Version:** 0.3 | **Family:** CONCORD | **Date:** March 2026

Purpose: This document defines the required outputs for CONCORD Scan. These schemas are the adoption wedge for existing backends and should be implementable even before runtime control components are introduced.

1. Scan bundle overview

Artifact	Required	Use
Readiness Report	Yes	Measures maturity across contract, consistency, authority, coordination, and recovery domains
Danger Map	Yes	Flags concrete failure modes and recommended patches
Resource Dependency Graph	Yes	Shows cross-resource mutation and consistency dependencies
Patch Plan	Recommended	Groups remediation steps into ordered work items
Generated Action Catalog	Recommended	Lists discovered and inferred actions

2. Readiness report schema

Field	Type	Meaning
scan_id	string	Unique scan identifier
target_name	string	Application or service name
generated_at	timestamp	Report generation time
overall_maturity_level	enum	Level 0 through Level 6
dimension_scores	object	Per-dimension score map
coverage_summary	object	Counts of resources, actions, endpoints, jobs, and validators
top_gaps	array	Highest-value missing controls
recommended_entry_sequence	array	Ordered adoption steps

- Dimension scores should cover: action contracts, state explicitness, consistency, idempotency, trust and budget, coordination, recovery, and observability.
- The report should distinguish discovered evidence from inferred evidence.

3. Danger map schema

Field	Type	Meaning
danger_id	string	Unique finding identifier
resource_or_endpoint	string	Affected resource, route, or job
danger_type	enum	Single-point or compound danger type
severity	enum	low, moderate, high, critical
trigger_condition	string	Condition that activates the risk
likely_failure_mode	string	Predicted operational impact
agent_risk	string	Why autonomous agents amplify the risk
recommended_patch	string	Suggested mitigation

3.1 Required single-point danger categories

- destructive endpoint without idempotency
- mutable resource without versioning
- shared resource without coordination
- implicit ordering dependency
- background job and API mutation collision
- missing authoritative recheck under eventual consistency
- unbounded bulk mutation path
- multi-step write flow without compensation coverage

3.2 Compound danger schema

Field	Type	Meaning
compound_danger_id	string	Unique compound finding identifier
contributing_risks	array	Referenced single-point risks
interaction_path	string	Workflow or dependency path where the risks combine
compound_severity	enum	Escalated severity score
silent_failure_likelihood	enum	low, moderate, high

Normative rule: Compound dangers shall be scored above any individual contributing risk when the combined path creates a credible silent corruption, double-commit, or unrecoverable drift scenario.

4. Resource dependency graph schema

Field	Type	Meaning
graph_id	string	Unique graph identifier
nodes	array	Resources, jobs, projections, and external dependencies
edges	array	Typed relationships among nodes
consistency_mismatches	array	Edges where consistency assumptions differ
saga_gaps	array	Multi-resource operations lacking saga coverage

hotspots	array	High-fan-in or high-fan-out nodes
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4.1 Required edge types

- mutates
- reads_before_mutation
- projects_to
- background_updates
- triggers
- depends_on_external

5. Patch plan schema

Field	Type	Meaning
patch_id	string	Recommended work item identifier
priority	enum	P0, P1, P2, P3
affected_assets	array	Files, routes, models, or jobs
expected_control_gain	string	What safety property is added
estimated_complexity	enum	small, medium, large
dependency_patch_ids	array	Predecessor work items

6. Scan implementation notes

- Scanner outputs must identify whether a field was discovered directly, inferred statistically, or derived heuristically.
- Where confidence is low, the scanner should emit a confidence band rather than a hard assertion.
- Every danger finding should be traceable to at least one source location, endpoint, migration, or runtime trace.
- Scanners should prefer visceral findings over abstract scoring in executive summaries.